

Tutorial 10 – Name Visibility & Functions cont.

Define all the functions inside a namespace.

example - Write a function in C++ to add two numbers by using call by reference method

```
#include <iostream>
using namespace std;
namespace example
{
    int add(int a, int b)
    {
        return a + b;
    }
}

int main()
{
    int result = example::add(5, 3);
    cout << result << endl;
    return 0;
}
```

1. Write a function in C++ to swap two numbers by using a call by reference method.
2. Write a function in C++ to calculate the area and perimeter of a rectangle. Pass length and width to the function using call by reference. Display the output within the function.
3. Write a function in C++ to modify an array. Function takes an array of integers and its size as parameters by reference, and modifies the array by doubling each element and displays it.
4. Write a function in C++ to reverse a string. Function takes a string parameter by reference and reverses it.
(
 use string header
 use str.length() calculate length of string
 for loop condition: i< half of string length
 swap characters using character index str(n - i - 1)
)

5. Write a C++ program using functions to find the volume of a cylinder ($v = \pi r^2 h$) and volume of a sphere ($v = \frac{4}{3} \pi r^3$).
- Use separate functions to find the volume of the cylinder and the volume of the sphere.
 - Declare all variables as global variables.
 - Pass variable values to the respective functions from main() using pass by reference.
 - Calculate and display answers within the functions.